

## Financial Coursework Report

Louis James Davidson

Student ID : 35712341

1(a)(i). To find the time when the bank loan is paid off, the following equations must be solved simultaneously:

$$140 \ddot{s}_{\overline{3}|0.05} = (3000p_1 - 120) \bar{a}_{\overline{t-3}|0.05} \quad (1)$$

$$50 \ddot{a}_{\overline{2}|0.035} = (3000p_1 - 120) \bar{s}_{\overline{43-t}|0.035} \quad (2)$$

From above, we acquire the following equation, which we solve via linear interpolation:

$$\frac{140 \ddot{s}_{\overline{3}|0.05}}{50 \ddot{a}_{\overline{2}|0.035}} = \frac{1 - 1.05^{3-t}}{1.035^{43-t} - 1} \implies t = 39.58953241 \text{ years} \quad (3)$$

1(a)(ii). Solving either (1) or (2) for  $p_1$  with the above value for  $t$ , we get:

$$(3000p_1 - 120) = \frac{22.6102159}{1 - 1.05^{3-t}} \implies p_1 = \pounds 0.04905598 \implies p_1 = 4.91 \text{ pence} \quad (4)$$

1(b)(i). We first find the new value for  $t$  using  $p_2 = \pounds 0.09811198$

$$140 \ddot{s}_{\overline{3}|0.05} = [3000p_2 - 120] \bar{a}_{\overline{t-3}|0.05} \implies t = 5.84708392 \text{ years} \quad (5)$$

Since the bank balance at time  $t = 0$  the sum of all cash flows after time  $t$  will be our final bank balance, so solving the below equation with the above  $p_1$  and  $t$  values:

$$Profit = (3000p_2 - 120)(1.035)^2 (\bar{s}_{\overline{43-t}|0.035}) - 50 \ddot{s}_{\overline{2}|0.035} \implies Profit = \pounds 13,954.1124 \text{ million} \quad (6)$$

1(c). For this model to work there are various assumptions we have made which include:

**The model.** To avoid circular references, the model approximates the time  $t$  at which the loan is fully repaid rather than solving for it exactly. An approximate value of  $t$  is derived from the bank-balance row, giving an error of about 0.006 years (roughly two days) for the base case, which is negligible compared with the 45-year expected project life. This trade-off allows the model to remain fully adaptable to changes in the input variables while maintaining an acceptable level of accuracy. Figure 2 further shows that our model is sufficiently accurate since the NPV predicted by the model follows an intuitive path, falling sharply at the start before slowly climbing just above 0 before falling back to 0 at the end (given an energy price of 0.049055988).

**Inflation.** We assume there is no inflation in the model, which is unrealistic over a 45-year time frame. In reality, operating costs (wages, fuel prices, energy price etc.) would increase year on year and reduce yearly profit unless the energy prices are increased accordingly. If prices cannot be fully adjusted for inflation, we can assume that the actual TNPV will be lower than the value predicted by the model. It can be noted that inflation may rise slower than energy costs - which would increase revenue more than operational costs. However it would not be wise to include this outcome in any presented forecasts, as such an event occurring is highly unlikely and unpredictable.

**Energy price.** In the base model, the energy price is assumed to be constant at 4.9p/kWh for the whole project. In practice, electricity prices are volatile and would change over time, driven by inflation, demand and government price-caps. We can see from figure 1, that even a small increase in the price from  $p = 0.0491$  results in the loan being paid off much sooner, this would imply that the project is highly risky if we expect the price to be near the break even price. It only takes a small reduction in the price of energy for the loan pay-off time to increase drastically or never be paid off. Power plants typically sell their energy onto the grid at an average price of 6.5p / kWh. If we use a cautious estimate of 4.9p/kWh the project does break even. To illustrate sensitivity however, if we instead assume a more optimistic average energy price of 6.5p/kWh, the model predicts a TNPV of approximately £727million and the loan is repaid about 7.35 years after construction finishes. This underlines that project value is highly sensitive to the assumed energy price.

**Construction.** We have assumed that construction time and costs are fixed, but in practice delays or

unexpected costs may happen for a variety of reasons, potentially extending the construction time by several years. This is one of the main risks of the project, as a delay in construction allows the loan to compound further while also delaying the time when profits start coming in. Delays would also extend the total time span of the project - which would make forecasting with accuracy even harder. Furthermore it is possible that a severe delay in construction could cause the loan to become too large for the bank to support - causing the project to not be finished, or sold off to other lenders.

**Efficiency and power production.** Over a 40-year operating life it is unrealistic to assume that the plant will maintain a constant annual output of 3B kWh. Ageing equipment, maintenance outages and operational constraints are likely to reduce output over time, which would directly affect annual revenue and therefore lower TNPV relative to the base-case estimate. In addition, the plant will not operate at 100% capacity at all times due to daily and seasonal fluctuations in demand. During the summer months a large share of the UK's electricity is supplied by renewable sources, an energy sector that is expected to grow over time. If these effects are not already captured in the assumed 3B kWh/year figure, the true TNPV of the project would be lower than in our base-case. The continuous payment stream used to model revenue also does not reflect seasonal patterns - more energy sold during the winter than in summer.

**Operating lifespan.** We have assumed that the plant operates for 40 years. In practice, the plant may wear faster than expected, leading to higher maintenance expenses or to premature closure, depending on whether it is economically viable to continue operating. Either outcome would reduce the TNPV of the project relative to the base case. The opposite may occur and the plant could be operated safely and economically beyond 40 years - extending the operational life and TNPV. Other factors such as net zero targets could also reduce the lifespan of the project - if government targets become stricter it may force the plant to close early.

**Operational costs.** The model treats annual operating costs as a constant continuous cashflow, but in practice they will fluctuate with time. Fuel prices, wages, maintenance and replacement of key components can all move unpredictably over a 40 year time frame, and unplanned outages or major equipment failures could lead to significant one-off repair costs. In addition, future regulatory or environmental requirements may impose extra operating expenses (e.g. enhanced monitoring or emissions control). Any sustained increase in operating costs would directly reduce annual profit, lowering TNPV relative to the base case.

**Borrowing rate.** We have assumed that the borrowing rate is fixed at 5% for the entire life of the project and use this as the single discount rate in the TNPV calculations. In reality, interest rates would fluctuate over time, driven by changes in the Bank of England base rate, inflation, the bank's own funding costs and the perceived risk of the project. Over a 45-year time frame it is impossible to forecast the path of rates with any precision, so a constant 5% should be viewed as a realistic assumption rather than a prediction. If the actual cost of borrowing is higher than 5%, the true TNPV would be lower than shown here, higher if the cost of borrowing is less than in our assumption. If we expect prices to be more favourable, it is possible to get more favourable terms from the bank, although this is entirely dependent on energy price forecasts.

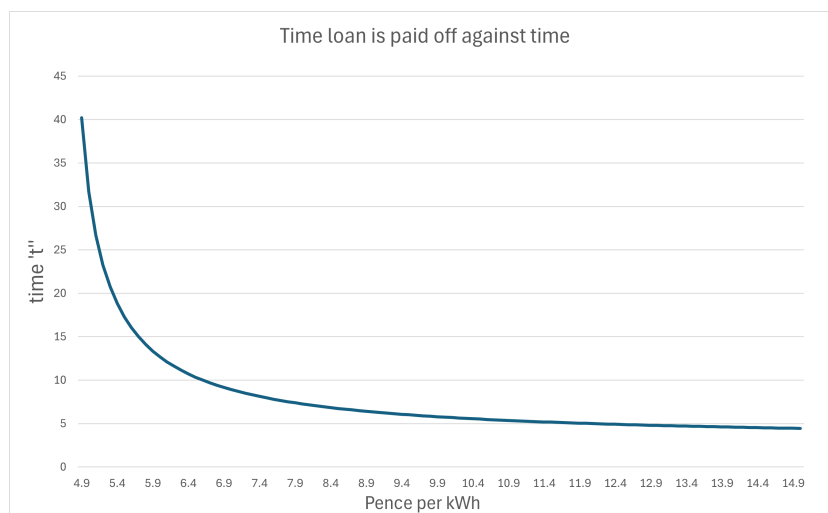


Figure 1: Time loan is paid of against energy price

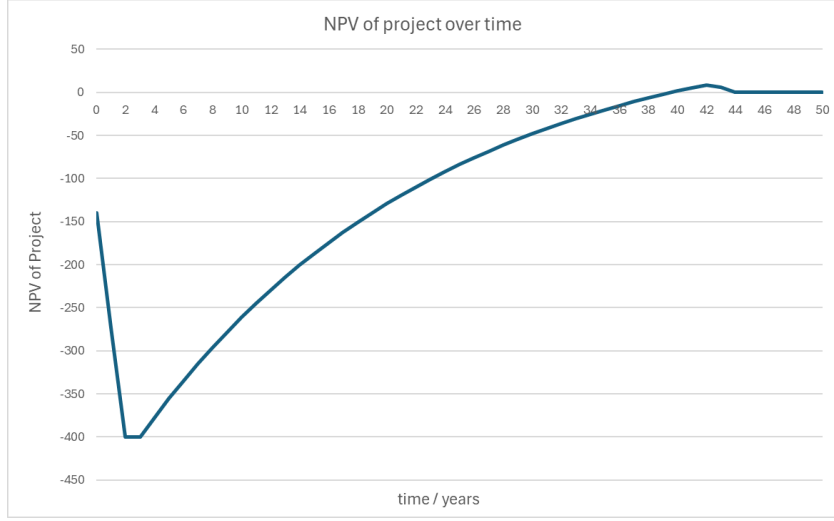


Figure 2: NPV of Project over Time

2(a).

$$E[X] = (1 + j)^3 \implies E[X] = \pounds 1.295029 \text{ million} \quad (7)$$

$$\sigma[X] = \sqrt{\text{Var}[X]} = \sqrt{[(1 + j)^2 + s^2]^3 - (1 + j)^6} = \pounds 0.10300075 \text{ million} \quad (8)$$

2(b).

$$E[Y] = \ddot{s}_{\overline{3}|j} = \pounds 3.573129 \text{ million} \quad (9)$$

and via the recursive formula:

$$\sigma[Y] = \pounds 0.208351433 \text{ million} \quad (10)$$

2(c).

$$\mu = \log \left( \frac{1 + j}{\sqrt{1 + \left( \frac{s}{1+j} \right)^2}} \right) = 0.0851267 \quad (11)$$

and

$$\sigma = \sqrt{\log \left( 1 + \left( \frac{s}{1+j} \right)^2 \right)} = 0.04584745 \quad (12)$$

2(d). We have

$$X = \prod_{t=1}^3 (1 + i_t) \implies \log X = \sum_{t=1}^3 \log(1 + i_t) \sim N(3\mu, 3\sigma^2). \quad (13)$$

Therefore the 95% confidence limits for the above normal distribution is:

$$\text{Lower CL} = e^{(3\mu - 1.96\sqrt{3\sigma^2})} \implies \text{Lower CL} = \pounds 1.10487948 \text{ million} \quad (14)$$

$$\text{Upper CL} = e^{(3\mu + 1.96\sqrt{3\sigma^2})} \implies \text{Upper CL} = \pounds 1.50836148 \text{ million} \quad (15)$$

2(e). Using 10,000 simulations we acquire estimates for:  $E[X]$ ,  $\sigma[X]$  and CL [X]

$$E[X] = \pounds 1.29283636 \text{ million} \quad , \quad \sigma[X] = \pounds 0.10300885 \text{ million} \quad (16)$$

$$\text{Lower CL} = \pounds 1.10500733 \text{ million} \quad , \quad \text{Upper CL} = \pounds 1.50674583 \text{ million} \quad (17)$$

Our estimated mean for X lies within the 95% confidence limits as expected. The difference between the calculated values and the estimations for  $E[X]$  and  $\sigma[X]$  are 0.00219 and  $8.105 \times 10^{-6}$  respectively, which are both negligible compared to the calculated values. Since the difference is so small, this suggests that

both of our calculated values are correct with a reasonable degree of certainty.

2(f). Using 10,000 simulations we acquire estimates for:  $E[Y]$  ,  $\sigma[Y]$  and CL [Y]

$$E[Y] = \pounds 3.56904661 \text{ million} , \sigma[Y] = \pounds 0.20911033 \text{ million} \quad (18)$$

$$\text{Lower CL} = \pounds 3.18037861 \text{ million} , \text{Upper CL} = \pounds 3.99741123 \text{ million} \quad (19)$$

Our estimated mean for Y lies within the 95% confidence limits as expected. The difference between the calculated values and the estimations for  $E[Y]$  and  $\sigma[Y]$  are 0.00408 and 0.00076 respectively, which are both negligible compared to the calculated values. Since the difference between calculated and estimated values is so small, this suggests that both of our calculated values are correct with a reasonable degree of certainty.

2(g). For  $k = 0.5$ , our estimated values for  $E[X]$  ,  $\sigma[X]$  and CL [X] are:

$$E[X] = \pounds 1.30147126 \text{ million} , \sigma[X] = \pounds 0.15003885 \text{ million} \quad (20)$$

$$\text{Lower CL} = \pounds 1.03536823 \text{ million} , \text{Upper CL} = \pounds 1.61955135 \text{ million} \quad (21)$$

For  $k = 1.0$ , our estimated values for  $E[X]$ ,  $\sigma[X]$  and CL [X] are:

$$E[X] = \pounds 1.31088805 \text{ million} , \sigma[X] = \pounds 0.22611491 \text{ million} \quad (22)$$

$$\text{Lower CL} = \pounds 0.92595311 \text{ million} , \text{Upper CL} = \pounds 1.80962149 \text{ million} \quad (23)$$

2(h). For  $k = 0.5$ , our estimated values for  $E[Y]$  ,  $\sigma[Y]$  and CL [Y] are:

$$E[Y] = \pounds 3.58379791 \text{ million} , \sigma[Y] = \pounds 0.30076161 \text{ million} \quad (24)$$

$$\text{Lower CL} = \pounds 3.04264194 \text{ million} , \text{Upper CL} = \pounds 4.20914783 \text{ million} \quad (25)$$

For  $k = 1.0$ , our estimated values for  $E[Y]$ ,  $\sigma[Y]$  and CL [Y] are:

$$E[Y] = \pounds 3.60068403 \text{ million} , \sigma[Y] = \pounds 0.47172659 \quad (26)$$

$$\text{Lower CL} = \pounds 2.76946120 \text{ million} , \text{Upper CL} = \pounds 4.60716472 \text{ million} \quad (27)$$

2(i). In order to make predictions about a stochastic model of an investment fund, we have assumed that the annual accumulation factor is i.i.d and lognormally distributed (for the independent case). This is strong assumption to make as in reality, no fund will follow an exact lognormal distribution and so the model can only ever approximate the behaviour experienced by the fund. Furthermore the annual accumulation factor is unlikely to be perfectly i.i.d due to a plethora of real world events influencing the annual accumulation factor from year to year.

To try to fix this we made a model that depends on the previous years annual accumulation factor. When adapting the model to one where the previous years returns effect the current years distribution, we see that expected values rise for both X and Y. This is not surprising as the additional factor  $\log(1+i)$  is on average larger than  $\mu$  . This increases the mean of the distribution of  $1+i$  , results in an increase to the expected value of the fund - as shown in the simulations. We see a more notable change when it comes to the s.d. of the model =, this is because the dependence on the previous year exaggerates shifts to either side of the mean - higher than expected gets even higher and vice versa. This results in a distribution with fatter tails and therefore a higher s.d.. As a result the annual accumulation value is more likely to take an extreme value - increasing the uncertainty of the model.